

Yorkinoy Shermatova

Postdoctoral Associate, Department of Ecology, Evolution and Behavior
University of Minnesota – Twin Cities

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RESEARCH INTERESTS

Mathematical and computational ecology; theoretical ecology and biodiversity; modeling of disease dynamics in forest ecosystems; climate mathematics; dynamical systems and individual / agent-based models; integration of remote sensing and machine learning with ecological models for forest health.

EDUCATION

Ph.D., Applied Mathematics December 2024

University of Minnesota – Twin Cities, USA

Minor: Ecology, Evolution, and Behavior

Thesis: *A Mathematical Exploration of Disease and Diversity in Oaks: The Case of Oak Wilt Disease*

Advisor: Prof. [Richard McGehee](#)

M.S., Mathematics June 2017

City College of New York, NY, USA

B.S., Mathematics July 2009

Samarkand State University, Uzbekistan

ACADEMIC APPOINTMENTS

Postdoctoral Associate Jan 2025 – Present

Department of Ecology, Evolution and Behavior, University of Minnesota – Twin Cities

Project: *Detecting Invasive Tree Diseases Using Spaceborne Phenological Observations* (MITPPC; PI: J. Pinto-Ledezma)

Graduate Instructor / Teaching Assistant Aug 2018 – Dec 2024

Department of Mathematics, University of Minnesota – Twin Cities

Graduate Research Assistant Summer 2022

College of Biological Sciences, University of Minnesota – Twin Cities (Oak Wilt modeling)

Adjunct Lecturer Aug 2017 – May 2018

Department of Mathematics, Fordham University, NY

Adjunct Lecturer Aug 2016 – May 2018

Department of Mathematics, City College of New York, NY

Research Assistant Jan 2013 – June 2015

Department of Mathematics, Samarkand State University, Uzbekistan

Spectral properties of Hamiltonian systems on lattices.

PUBLICATIONS

Submitted Manuscripts

Shermatova, Y., McGehee, R., & Cavender-Bares, J. (2026). *Mathematical modeling of oak wilt dynamics: The protective role of white oaks in mixed forests*. Submitted to *Ecological Modelling*.

Manuscripts in Preparation

Shermatova, Y., & Pinto-Ledezma, J. *Primed for epidemic: Land-use legacies as drivers of oak wilt emergence in eastern North America*. (Perspective; in preparation.)

Peer-Reviewed Journal Articles

Muminov, M. E., & Shermatova, Y. M. (2016). On finiteness of the discrete spectrum of a three-particle Schrödinger operator on a lattice. *Russian Mathematics (Iz. VUZ)*, 60(1), 22–29. [\[doi\]](#)

Dell’Antonio, G. F., Muminov, Z. I., & Shermatova, Y. M. (2011). On the number of eigenvalues of a model operator related to a system of three particles on lattices. *Journal of Physics A: Mathematical and Theoretical*, 44(31), 315302. [\[doi\]](#)

Conference Proceedings

Shermatova, Y. (2013). In *Proceedings of the Scientific Republic Conference: Current Problems of Complex Analysis*, Tashkent, Uzbekistan.

GRANTS AND RESEARCH FUNDING

Co-Lead. *Building Bridges & Biodiversity: Empowering the Commonwealth Terrace Cooperative Community*. July 2025 – June 2026

Institute on the Environment (IonE) Spring 2025 Mini Grant, University of Minnesota. Award: \$3,000. PI: Jesús Pinto-Ledezma; Co-Leads: Y. Shermatova & K. Pathirana. Community partner: Commonwealth Terrace Cooperative.

Postdoctoral Researcher (Funded). *Detecting Invasive Tree Diseases Using Spaceborne Phenological Observations*. 2024 – ongoing

Minnesota Invasive Terrestrial Plants and Pests Center (MITPPC), University of Minnesota. Total award: \$435,937.

PI: Jesús Pinto-Ledezma; Postdoctoral Associate: Y. Shermatova. [\[project page\]](#)

Affiliated Researcher. *Detection and Treatment of Oak Wilt*.

Minnesota Invasive Terrestrial Plants and Pests Center (MITPPC), University of Minnesota. [\[project page\]](#)

Graduate Research Funding. Oak wilt disease modeling.

Summer 2022

College of Biological Sciences, University of Minnesota.

HONORS AND AWARDS

• **Ella Thorp Graduate Fellowship**, Department of Mathematics, University of Minnesota Spring 2024

• **COGS Outstanding Teaching Assistant Award**, University of Minnesota 2020–2021

INVITED TALKS

• *Modeling Oak Wilt Spread Using Individual-Based Models*.

Pacific Northwest SIAM Biennial Conference, Bellingham, WA.

Oct 2023

CONTRIBUTED TALKS AND CONFERENCE PRESENTATIONS

- *Individual-Based Modeling to Understand Oak Wilt Disease Spread*. SIAM Conference on Applications of Dynamical Systems (DS23), Portland, OR. 2023
- *Planetary Health: Advancing Climate Solutions Now*. IonE Ecosystem Health Cohort, University of Minnesota. 2023
- *Oak Wilt Disease and Species Coexistence*. ASCEND Winter Symposium. 2022
- *Agent-Based Models and Their Applications*. UMN Math Climate Seminar. 2022
- *Disease and Fire Interaction*. UMN Math Climate Seminar. 2022
- *Polar Vortex*. SIAM Conference on Mathematics of Planet Earth (MPE20). 2020
- *Inter-Country Variability of COVID-19 Case Fatality Rates*. UMN Math Climate Seminar. 2020

CONFERENCE AND SESSION ORGANIZATION

- **Co-Convener**, Session B020: *Bridging Remote Sensing, Machine Learning, and Ecological Modeling to Address Forest Health Challenges*. American Geophysical Union (AGU) Annual Meeting, New Orleans, LA. Dec 2025
- **Session Chair**, Graduate Student Combinatorics Conference, University of Minnesota. April 2021

TEACHING EXPERIENCE

Instructor of Record

University of Minnesota – Twin Cities 2020 – 2024

- MATH 1371 – Calculus I (College of Science and Engineering), Fall 2024
Enrollment: 32. Student Rating of Teaching: **instructor mean 5.73 / 6.00**; “well prepared” 5.85, “treated me with respect” 5.85, “feedback to improve performance” 5.77, “would recommend instructor” 5.69.
- Calculus I – multiple sections (College of Science and Engineering)
- Precalculus II – multiple sections

Across courses, enrollments ranged from 32 to 86; student rating means consistently exceeded 5.0/6.0.

Fordham University 2017 – 2018

- Finite Mathematics (for Business majors)
- Applied Calculus (for Business majors)

City College of New York 2016 – 2018

- College Algebra; Trigonometry; Precalculus; Calculus

Teaching Assistantships

- University of Minnesota – Twin Cities: undergraduate mathematics courses, including writing-intensive sequences. 2018 – 2024

Curriculum Development

- Co-developer, Precalculus curriculum revision, with Dr. Mike Weimerskirch (UMN). Summer 2022
- Revised MATLAB lab assignments for Linear Algebra and Differential Equations, with Dr. Julie Leifeld (UMN). Summer 2024

MENTORING

- Graduate mentor, [Mathematics Directed Reading Program \(DRP\)](#), University of Minnesota – mentored an undergraduate student in independent study.

PROFESSIONAL SERVICE

- Co-Convener, Session B020, AGU Annual Meeting (see *Conference and Session Organization*). 2025
- Member, Professional Development Committee, Gender Minorities in Mathematics, University of Minnesota. Oct 2023 – Nov 2024
- Counselor, Mathematics Project at Minnesota (MPM), University of Minnesota. Jan 2022
- Session Chair, Graduate Student Combinatorics Conference, University of Minnesota. April 2021

COMMUNITY ENGAGEMENT AND OUTREACH

- Board Member, Commonwealth Terrace Cooperative (CTC), University of Minnesota. Nov 2023 – Present
- Programs Chair and Yoga Instructor, Commonwealth Terrace Cooperative (CTC). April 2023 – Present

WORKSHOPS AND SUMMER SCHOOLS ATTENDED

- Math Climate Research Network (MCRN) Summer School – Polar Vortex working group. 2019, 2020

PROFESSIONAL AFFILIATIONS

- Math Climate Research Network (MCRN)
- Advancing Spectral Biology in Changing Environments to Understand Diversity (ASCEND)
- Association for Women in Mathematics (AWM)
- American Geophysical Union (AGU)

TECHNICAL SKILLS

Programming & Analysis: Python, MATLAB

Tools: $\text{\LaTeX}$, Git

Modeling: Individual-based / agent-based models, dynamical systems, ordinary and partial differential equations, statistical analysis

LANGUAGES

English (professional fluency) · Russian (fluent) · Uzbek (native)

REFERENCES

Available upon request.